CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 8: Profit, Loss and Discount

Topic-wise short notes with solved examples

Cost price, selling price, profit and loss as amounts and percentages, overhead expenses, marked price and discount, successive discounts and tax on sales.

1 Basic terms

- **Cost price (C.P.)**: the price at which an article is bought. **Selling price (S.P.)**: the price at which it is sold.
- **Overheads**: extra expenses such as transport, repairs and labour. They are **added to the cost price**.
- **Profit (gain)** $=$ S.P. $-$ C.P., when S.P. $>$ C.P.
- **Loss** $=$ C.P. $-$ S.P., when C.P. $>$ S.P.
- Profit and loss percentages are always calculated **on the cost price**.

$$\text{Profit}\% = \frac{\text{Profit}}{\text{C.P.}} \times 100, \qquad \text{Loss}\% = \frac{\text{Loss}}{\text{C.P.}} \times 100$$

Example 1.1 A table bought for ₹1600 is sold for ₹1840. Find the profit per cent.

SOLUTION

Profit $= 240.\ \frac{240}{1600} \times 100$

15%

Example 1.2 A man buys a second-hand cycle for ₹2400, spends ₹300 on repairs and sells it for ₹2430. Find his loss per cent.

SOLUTION

Total C.P. $= 2700.$ Loss $= 270.\ \frac{270}{2700} \times 100$

10%

2 Finding S.P. and C.P.

To find	Given a profit of $p\%$	Given a loss of $l\%$
S.P.	C.P. $\times \frac{100+p}{100}$	C.P. $\times \frac{100-l}{100}$
C.P.	S.P. $\times \frac{100}{100+p}$	S.P. $\times \frac{100}{100-l}$

Example 2.1 A fan costs ₹1500. At what price must it be sold to gain 12%?

SOLUTION

$1500 \times \frac{112}{100}$

₹1680

Example 2.2 By selling a watch for ₹1140 a shopkeeper loses 5%. Find the cost price.

SOLUTION

$1140 \times \frac{100}{95}$

₹1200

Example 2.3 By selling an article for ₹720 a man loses 10%. At what price should he sell it to gain 5%?

SOLUTION

C.P. $= 720 \times \frac{100}{90} = 800$. New S.P. $= 800 \times \frac{105}{100}$.

₹840

3 Several articles

- When the C.P. of x articles equals the S.P. of y articles, take the C.P. of one article as ₹1.
- When articles are bought at one rate and sold at another, find the C.P. and S.P. of the **same number** of articles (use the LCM).
- Two articles sold at the **same price**, one at a gain of $x\%$ and the other at a loss of $x\%$: there is always a **loss of** $\frac{x^2}{100}\%$ overall.

Example 3.1 The cost price of 12 pens equals the selling price of 10 pens. Find the profit per cent.

SOLUTION

Let the C.P. of one pen be ₹1. S.P. of 10 pens $= 12$, C.P. of 10 pens $= 10$.

$\frac{2}{10} \times 100 = 20\%$ **profit**

Example 3.2 Lemons are bought at 5 for ₹20 and sold at 4 for ₹20. Find the gain per cent.

SOLUTION

C.P. of one $= 4$; S.P. of one $= 5$.

25%

Example 3.3 A man sells two radios for ₹1980 each, gaining 10% on one and losing 10% on the other. Find his overall gain or loss.

SOLUTION

C.P. of the first $= 1980 \times \frac{100}{110} = 1800$; of the second $= 1980 \times \frac{100}{90} = 2200$.

Total C.P. $= 4000$; total S.P. $= 3960$.

A loss of ₹40, that is 1%

4 Discount

- **Marked price (M.P.)** or list price: the price printed on the article.
- **Discount**: a reduction offered on the marked price. Discount is always calculated **on the marked price**.
- S.P. $=$ M.P. $-$ discount $=$ M.P. $\times \frac{100 - d}{100}$, where d is the discount per cent.

$$\text{Discount}\% = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

Example 4.1 A shirt marked ₹800 is sold at a discount of 15%. Find the selling price.

SOLUTION

$800 \times \frac{85}{100}$

₹680

Example 4.2 After a discount of 20%, a bag is sold for ₹960. Find its marked price.

SOLUTION

$960 \times \frac{100}{80}$

₹1200

Example 4.3 An article marked ₹1250 is sold for ₹1100. Find the discount per cent.

SOLUTION

$\frac{150}{1250} \times 100$

12%

5 Marked price, discount and profit together

- First get the S.P. from the M.P. and the discount; then compare the S.P. with the C.P.
- To fix the marked price: find the required S.P. from the C.P. and profit, then divide by $\frac{100 - d}{100}$.

Example 5.1 A shopkeeper buys an article for ₹600 and marks it 30% above the cost price. He allows a discount of 10%. Find his profit per cent.

SOLUTION

M.P. $= 780$. S.P. $= 780 \times 0.9 = 702$. Profit $= 102$.

17%

Example 5.2 How much above the cost price must a trader mark his goods so that after allowing a discount of 20% he still gains 12%?

SOLUTION

Let C.P. $= 100$. Required S.P. $= 112$. M.P. $= 112 \times \dfrac{100}{80} = 140$.

40% above the cost price

6 Successive discounts

- Two discounts of $d_1\%$ and $d_2\%$ one after the other: S.P. $=$ M.P. $\times \dfrac{100 - d_1}{100} \times \dfrac{100 - d_2}{100}$.
- They are equivalent to a single discount of $\left(d_1 + d_2 - \dfrac{d_1 d_2}{100}\right)\%$, which is less than $d_1 + d_2$.

Example 6.1 Find the single discount equivalent to successive discounts of 20% and 10%.

SOLUTION

$20 + 10 - \dfrac{200}{100}$

28%

Example 6.2 A washing machine marked ₹15,000 is sold with successive discounts of 10% and 5%. Find the selling price.

SOLUTION

$15000 \times 0.9 \times 0.95$

₹12,825

7 Tax on sales

- A tax such as GST is charged as a percentage of the **selling price** and added to the bill.
- Amount paid $=$ S.P. $\times \left(1 + \dfrac{\text{tax}\%}{100}\right)$.
- If the price **includes** tax, the price before tax $=$ amount paid $\div \left(1 + \dfrac{\text{tax}\%}{100}\right)$.

Example 7.1 A pair of shoes is priced at ₹1500 and the tax is 12%. Find the amount to be paid.

SOLUTION

1500×1.12

₹1680

Example 7.2 A fan is bought for ₹2360 including tax at 18%. Find its price before tax.

SOLUTION

$2360 \div 1.18$

₹2000

8 Common mistakes

- Calculating profit% or loss% on the selling price.
- Calculating discount on the cost price instead of the marked price.
- Leaving out the overheads from the cost price.
- Adding successive discounts: 20% and 10% make 28%, not 30%.
- Using $100 + l$ for a loss or $100 - p$ for a profit.
- Finding the C.P. by taking the percentage of the S.P.

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